

(Theoretical Analysis) Thanks for this suggestion. In the revision, we provide a theoretical analysis demonstrating that **sink divergence is proportional to the scalar projection of the harmful vector (the negative direction of the safety vector)**, as shown below.

Formal Notation and Definitions. To further explore the intuition behind sink divergence, we analyze it in conjunction with **safety vectors** [1]. A safety vector represents a safety-related direction in the hidden space that captures systematic differences between refusal and harmful behaviors. Following the formulation in Eq. (7), we define the **harmful vector** for each attention head h as the average difference in activations between the harmful dataset X_m and the refusal dataset X_r :

$$\mathbf{v}_h = \frac{1}{N} \sum_{i=1}^N (\mathbf{c}_h(X_{m,i}) - \mathbf{c}_h(X_{r,i})), \quad (1)$$

where $X_{m,i} \in X_m$ and $X_{r,i} \in X_r$ denote paired harmful and refusal inputs, and $\mathbf{c}_h(\cdot)$ denotes the hidden activation extracted at the h -th head.

Theorem.

The sink divergence d_h of head h is proportional to the scalar projection of the harmful vector $\mathbf{v}_h$ onto the sink subspace:

$$d_h \approx \frac{1}{\|\mathbf{v}_{sink}\|} \text{Proj}_{\mathbf{v}_{sink}}(\mathbf{v}_h). \quad (2)$$

Proof.

First, using the attention sink mechanism, we decompose the head's activation $\mathbf{c}_h(X)$ into a sink component and a semantic residual component:

$$\mathbf{c}_h(X) = \alpha_h(X) \mathbf{v}_{sink} + \mathbf{r}_h(X), \quad (3)$$

where $\alpha_h(X)$ is the sink value, $\mathbf{r}_h(X)$ is the residual semantic vector, and $\mathbf{v}_{sink} \in \mathbb{R}^{d_k}$ represents the value vector associated with the sink token position (typically the first token, e.g., `<S>`). Substituting this decomposition into the definition of the harmful vector $\mathbf{v}_h$:

$$\mathbf{v}_h = \frac{1}{N} \sum_{i=1}^N [(\alpha_h(X_{m,i}) - \alpha_h(X_{r,i})) \mathbf{v}_{sink} + (\mathbf{r}_h(X_{m,i}) - \mathbf{r}_h(X_{r,i}))]. \quad (4)$$

By distributing the summation and factoring out $\mathbf{v}_{sink}$, we obtain:

$$\mathbf{v}_h = \underbrace{\left[\frac{1}{N} \sum_{i=1}^N (\alpha_h(X_{m,i}) - \alpha_h(X_{r,i})) \right]}_{d_h} \mathbf{v}_{sink} + \underbrace{\frac{1}{N} \sum_{i=1}^N (\mathbf{r}_h(X_{m,i}) - \mathbf{r}_h(X_{r,i}))}_{\Delta \mathbf{r}_h}. \quad (5)$$

The term in the first bracket is precisely the definition of sink divergence d_h (as per Eq. 3). Thus, the relationship simplifies to:

$$\mathbf{v}_h = d_h \mathbf{v}_{sink} + \Delta \mathbf{r}_h. \quad (6)$$

To compute the scalar projection of $\mathbf{v}_h$ onto the sink direction, we take the inner product with the unit sink vector $\hat{\mathbf{v}}_{\text{sink}} = \frac{\mathbf{v}_{\text{sink}}}{\|\mathbf{v}_{\text{sink}}\|}$:

$$\langle \mathbf{v}_h, \hat{\mathbf{v}}_{\text{sink}} \rangle = d_h \|\mathbf{v}_{\text{sink}}\| + \langle \Delta \mathbf{r}_h, \hat{\mathbf{v}}_{\text{sink}} \rangle. \quad (7)$$

Since attention sinks primarily capture structural placeholders rather than semantic shifts [2][3], the semantic difference $\Delta \mathbf{r}_h$ is approximately orthogonal to the sink subspace, i.e., $\langle \Delta \mathbf{r}_h, \hat{\mathbf{v}}_{\text{sink}} \rangle \approx 0$. Therefore, the sink divergence can be expressed as:

$$d_h = \frac{\langle \mathbf{v}_h, \hat{\mathbf{v}}_{\text{sink}} \rangle}{\|\mathbf{v}_{\text{sink}}\|} = \frac{1}{\|\mathbf{v}_{\text{sink}}\|} \text{Proj}_{\mathbf{v}_{\text{sink}}}(\mathbf{v}_h). \quad (8)$$

In summary, sink divergence is proportional to the scalar projection of the harmful vector $\mathbf{v}_h$ onto the sink subspace. Therefore, it provides an effective scalar proxy for quantifying harmful representations. This result further supports the validity and effectiveness of our proposed *Surgery* method.

(Result of theoretical analysis) In the theoretical analysis, we derive that the sink divergence d_h is proportional to the scalar projection of the harmful vector $\mathbf{v}_h$ onto the sink subspace, i.e., $d_h \approx \frac{1}{\|\mathbf{v}_{\text{sink}}\|} \text{Proj}_{\mathbf{v}_{\text{sink}}}(\mathbf{v}_h)$.

References

- [1] Arditi A, Obeso O, Syed A, et al. Refusal in language models is mediated by a single direction. *Advances in Neural Information Processing Systems*, 2024, 37: 136037–136083.
- [2] Xiao G, Tian Y, Chen B, et al. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023.
- [3] Guo T, Pai D, Bai Y, et al. Active-dormant attention heads: Mechanistically demystifying extreme-token phenomena in LLMs. *arXiv preprint arXiv:2410.13835*, 2024.